

External Libraries

about external libraries and we're going to finish the checkout page in our final

External Libraries

= code that is outside of our project

```
20
```

```
cart.forEach((cartItem) => {
```

```
20
```

```
cart.forEach((cartItem
```

Other developers



Internet

JavaScript code and then put their code on the internet and we can actually load

Our project

Lesson 15

Hello external library: <https://unpkg.com/supersimpledev@1.0.1/hello.js>

DayJS external library: <https://unpkg.com/dayjs@1.11.10/dayjs.min.js>

Hello (ESM version): <https://unpkg.com/supersimpledev@1.0.1/hello.esm.js>

DayJS (ESM version): <https://unpkg.com/supersimpledev@8.5.0/dayjs/esm/index.js>

How to Put a Website on the Internet: <https://youtu.be/p1QU3kLFPdg>

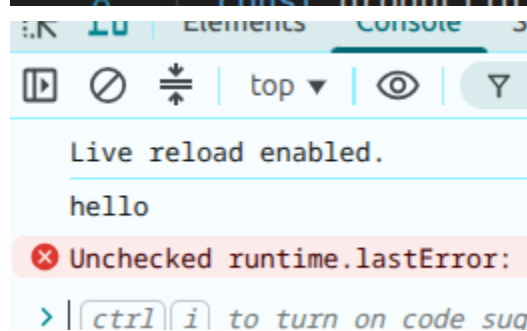
```
function hello() {  
  console.log('hello');  
}
```

**External library
= code that is outside of our project**

bottom and above the script tag we're
going to add

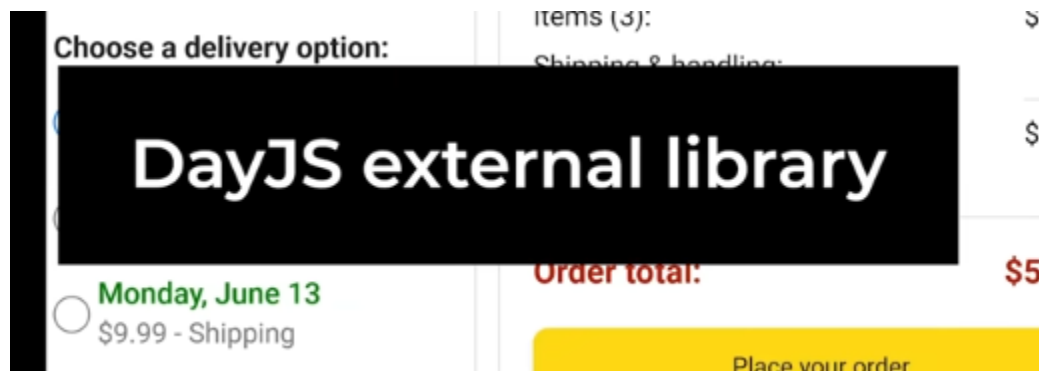
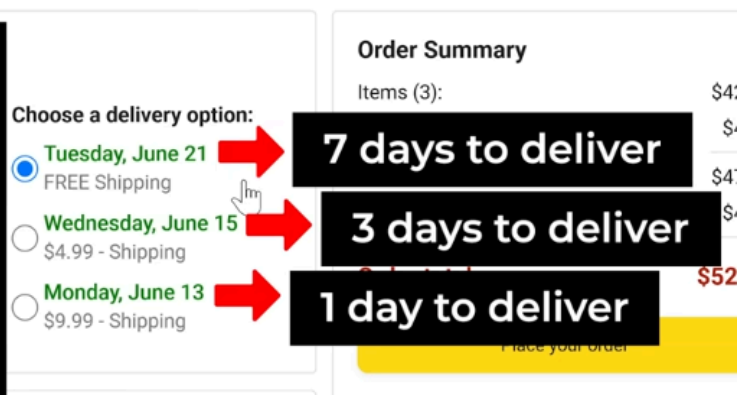
going to give this a URL of a website
that contains JavaScript

```
JS checkout.js M x checkout.html M JS cart.js
AmazonProject > scripts > JS checkout.js > ...
1 import{cart, removeFromCart} from '../data/cart.js'
2 import{products} from '../data/products.js'
3 import { formatCurrency } from '../utils/formatCurrency.js'
4
5 | hello();
6 let cartSummaryHTML = '';
7 cart.forEach((cartItem)=>{
8   const productId = cartItem.productId;
```



To get these dates:

1. Get today's date
2. Do calculations (Add 7 days, ...)
3. Display the date in easy-to-read format



```

1,4}|M{1,4}|D{1,2}|d{1,4}|H{1,2}|h{1,2}|a|A|m{1,2}|s{1,2}|Z{1,2}|SSS/g,M=
ekdays:"Sunday_Monday_Tuesday_Wednesday_Thursday_Friday_Saturday".split("_"),months:"January_February_March_April_May_
ber_October_November_December".split("_"),ordinal:function(t){var e=["th","st","nd","rd"],n=t%100;return "["+t+(e[(n-
)|e[0]]+"")+"]"},m=function(t,e,n){var r=String(t);return!r||r.length>=e?t:""+Array(e+1-r.length).join(n)+t},v={s:m,z:fu
Offset(),n=Math.abs(e),r=Math.floor(n/60),i=n%60;return(e<=0?"-":"")+m(r,2,"0")+":"+m(i,2,"0")},m:function t(e,n){if(
urn-t(n,e);var r=12*(n.year()-e.year())+(n.month()-e.month()),i=e.clone().add(r,c),s=n-i<0,u=e.clone().add(r+(s?-1:1).
?i-u:-i)||0},a:function(t){return t<0?Math.ceil(t)||0:Math.floor(t)},p:function(t),u:function(t){return void
},w=function t(e,n,r){var i;if(
&&u.length>1)return t(u[0])}else
ject"==typeof e?e:{};return
e.$u,x:e.$x,$offset:e.$offset}}},
p=!0}var m=M.prototype;return
function M(t){this.$L=w(t.locale
ion(t){this.$d=function(t){var e=t.date,n=t.utc;if(!n)return new Date(nan);if(!u(e))return new Date;if(e instai
ew Date(e);if("string"==typeof e&&!/$$/i.test(e)){var r=e.match($);if(r){var i=r[2]-1||0,s=(r[7]||"0").substring(0,3).
.UTC(r[1],i,r[3]||1,r[4]||0,r[5]||0,r[6]||0,s)):new Date(r[1],i,r[3]||1,r[4]||0,r[5]||0,r[6]||0,s)}}return new Date(e
()),m.init=function(){var
s.$y=t.getFullYear(),this.$M=t.getMonth(),this.$D=t.getDate(),this.$W=t.getDay(),this.$H=t.getHours(),this.$m=t.getMir
onds(),this.$ms=t.getMilliseconds(),m.$utils=function(){return b},m.isValid=function(){return!
ring()===1}}m.isSame=function(t,e){var n=0;return this.startOf(e)<=n&&n<=this.endOf(e)}m.isAfter=function(t,e){ru

```

Minification

Creates a function daysjs()

```

hello.js:2
console.log(daysjs());
let cartSummaryHTML = ''

```

Show console sidebar

hello.js:2

checkout.js:6

► M {\$L: 'en', \$d: Sun Jul 05 2026 07:55:07 GMT+0600 (Bangladesh Standard Time), \$y: 2026, \$M: 6, \$D: 5, ...}

Unchecked runtime.lastError: The message port closed before a response was received.

checkout.html:1

Calculate delivery date:

1. Get today's date
2. Do calculations
(Add 7 days, ...)
3. Display the date in
easy-to-read format

Add

Returns a cloned Day.js object with a specified amount of time added.

```
const a = dayjs()  
const b = a.add(7, 'day')
```



```
6  
7 const today = dayjs();  
8 const deliveryDate = today.add(7, 'days');  
9
```

```
6  
7 const today = dayjs();  
8 const deliveryDate = today.add(7, 'days');  
9 deliveryDate.format('M');
```

List of all available formats

Format	Output	Description
YY	18	Two-digit year
YYYY	2018	Four-digit year
M	1-12	The month, beginning at 1
MM	01-12	The month, 2-digits

dd	Su-Sa	The min name of the day of the week
ddd	Sun-Sat	The short name of the day of the week
dddd	Sunday-Saturday	The name of the day of the week
H	0-23	The hour

```
8 const deliveryDate = today.add(7, 'days');  
9 console.log(deliveryDate.format('dddd, MMMM D'));  
0
```

hello

Sunday, July 12

Best Practice

When we need something complicated,

- Try to find an external library first.
- Before writing the code ourselves.

How to find external libraries



🔍 javascript date library



External Libraries + JavaScript Modules

ESM Version

ESM = EcmaScript Module

ESM Version

A version that works with
JavaScript Modules

```
export function hello() {  
  console.log('hello');  
}
```

```
money.js ,  
import {hello} from 'https://unpkg.com/  
supersimpledev@1.0.1/hello.esm.js';
```

```
hello();
```

```

money.js ,
import {hello} from 'https://unpkg.com/
supersimpledev@1.0.1/hello.esm.js';
import dayjs from 'https://unpkg.com/
supersimpledev@8.5.0/dayjs/esm/index.js';

hello();

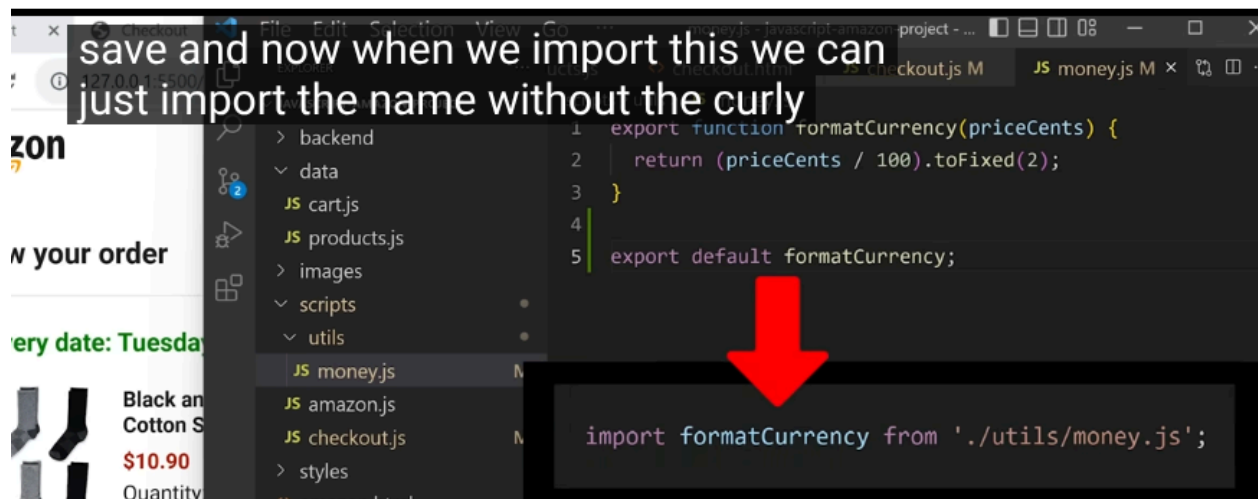
const today = dayjs();
const deliveryDate = today.add(7, 'days');
console.log(deliveryDate.format('dddd, MMMM
D'));

```

Default Export

- another way of exporting
- we can use it when we only want to export 1 thing

save and now when we import this we can just import the name without the curly



```

1 export function formatCurrency(priceCents) {
2   return (priceCents / 100).toFixed(2);
3 }
4
5 export default formatCurrency;

```

```

import formatCurrency from './utils/money.js';

```

Each file can only have
1 default export

curly brackets is called a named
export

Named Export

review your order

```
cart = [{  
  productId: 'e43638ce-6aa0',  
  quantity: 2,  
}, {  
  productId: '15b6fc6f-327a',  
  quantity: 1,  
}]
```

Choose a delivery option:

- ☒ Tuesday, June 21
FREE Shipping
- ☐ Wednesday, June 15
\$4.99 - Shipping
- ☐ Monday, June 13
\$9.99 - Shipping

Order Summary

1. Delivery time

Total before tax:

(10%):

2. Price

Order total:

Place your order

```
cart = [{  
  productId: 'e43638ce-6aa0',  
  quantity: 2,  
  deliveryTime: '3 days',  
  deliveryPrice: 499  
}, {  
  productId: '15b6fc6f-327a',  
  quantity: 1,  
  deliveryTime: '3 days',  
  deliveryPrice: 499  
}]
```

Choose a delivery option:

- ☒ Tuesday, June 21
FREE Shipping
- ☐ Wednesday, June 15
\$4.99 - Shipping
- ☐ Monday, June 13
\$9.99 - Shipping

Order Summary

1. Delivery time

Total before tax:

(10%):

2. Price

Order total:

Place your order

```
deliveryOptions = [{
  id: '1',
  deliveryTime: '7 days',
  deliveryPrice: 0
}, {
  id: '2',
  deliveryTime: '3 days',
  deliveryPrice: 499
}]
```

```
cart = [{
  productId: 'e43638ce-6aa0',
  quantity: 2,
  deliveryOptionId: '2'
}, {
  productId: '15b6fc6f-327a',
  quantity: 1,
  deliveryOptionId: '2'
}]
```

Choose a delivery option:

- ☒ Tuesday, June 21
FREE Shipping
- ☐ Wednesday, June 15
\$4.99 - Shipping
- ☐ Monday, June 13
\$9.99 - Shipping

```
deliveryOptions = [{
  id: '1',
  deliveryTime: '7 days',
  deliveryPrice: 0
}, {
  id: '2',
  deliveryTime: '3 days',
  deliveryPrice: 499
}]
```

Normalizing the Data

sense 999 and

Checkout (3 items)

Choose a delivery option:

- ☒ Tuesday, June 21
FREE Shipping
- ☐ Wednesday, June 15
\$4.99 - Shipping
- ☐ Monday, June 13
\$9.99 - Shipping

```
data > deliveryOptions.js
1  const deliveryOptions = [{
2    id: '1',
3    deliveryDays: 7,
4    priceCents: 0
5  }, {
6    id: '2',
7    deliveryDays: 3,
8    priceCents: 499
9  }, {
10   id: '3',
11   deliveryDays: 1,
12   priceCents: 999
13 }];
```

```
if (!cart) {
  cart = [{
    productId:
      'e43638ce-6aa0-4b85-b27f-e1d07eb678c6',
    quantity: 2,
    deliveryOptionId: '1'
  }, {
    productId:
      '15b6fc6f-327a-4ec4-896f-486349e85a3d',
    quantity: 1,
    deliveryOptionId: '2'
  }];
}
```

```

if(!cart){ // this is the previous code
  cart = [{
    productId:
      'e43638ce-6aa0-4b85-b27f-e1d07eb678c6',
    quantity :2
  }],
{
  productId
    : '5b6fc6f-327a-4ec4-896f-486349e85a3d',
  quantity: 1
}];
}

```

▼ JAVASCRIPT

▼ AmazonProject

▼ backend

{ } products.json

▼ data

JS cart.js M

JS deliveryOptio... U

JS products.js

> images

▼ scripts

▼ utils

JS money.js

JS amazon.js

JS checkout.js

AmazonProject > data > JS deliveryOptions.js > deliveryOption

```

1  const deliveryOptions = [{
2    id : '1',
3    deliveryDates : 7,
4    pricecents : 0
5  },{
6    id = '2',
7    deliveryDates : 3,
8    pricecents: 499,
9    pricecents : 499
10 },{
11   id: '3',
12   deliveryDates: 1,
13   pricecents: 999
14 }]

```

```

if(!cart){ // integrate delivery with
normalization
  cart = [{
    productId:
      'e43638ce-6aa0-4b85-b27f-e1d07eb678c6',
    quantity :2,
    deliveryOptionId: '1'
  },
  {
    productId
      : '5b6fc6f-327a-4ec4-896f-486349e85a3d',
    quantity: 1,
    deliveryOptionId: '2'
  }
];

```

```

    if(matchingItem){
      matchingItem.quantity +=1;
    }
    else{
      cart.push ({
        productId: productId, // use product
        instead of productName
        quantity : 1,
        deliveryOptionId: '1'
      });
    }
  }
  saveToStorage();
};

```

✖ ▶ Uncaught (in promise) TypeError: Cannot read properties of undefined (reading 'deliveryDays')
 at [orderSummary.js:23:22](#)
 at Array.forEach (<anonymous>)
 at renderOrderSummary ([orderSummary.js:12:8](#))
 at loadPage ([checkout.js:26:3](#))

> localStorage.clear()

```
<div class="delivery-options">
  <div
    class="delivery-options-title">
    Choose a delivery option:
  </div>
  <div class="delivery-option">
    <input type="radio" checked|
      class="delivery-option-input"
      name="delivery-option-$
        {matchingProduct.id}">
    <div>
```

Steps

1. Loop through deliveryOptions
2. For each option,
generate some HTML
3. Combine the HTML together

```
function deliveryOptionsHTML() {
  deliveryOptions.forEach((deliveryOption) => {
    const today = dayjs();
    const deliveryDate = today.add(
      deliveryOption.deliveryDays,
      'days'
    );
    const dateString = deliveryDate.format(
      'dddd, MMMM D'
    );

    const priceString = deliveryOption.priceCents
    === 0
    ?
    :
  });
}
```

only want it to be checked if it matches
the delivery option ID

the delivery option ID that is saved in
the cart so let's

```
function deliveryOptionHTML(matchingProduct, cartItem) {
  let html = '';

```

```
    Choose a delivery option:
  </div>
  ${deliveryOptionHTML(matchingProduct, cartItem)}
</div>
```

```
let cartSummaryHTML = '';
cart.forEach((cartItem) => {
  const productId = cartItem.productId;
```

```
const isChecked = deliveryOption.id === cartItem.deliveryOptionId;
```

```
html += `
```

```
productId:  
'e43638ce-6aa0-4b85-b27f'  
quantity: 2,  
deliveryOptionId: '1'
```

```
}, {  
productId:  
'15b6fc6f-327a-4ec4-896f'  
quantity: 1,  
deliveryOptionId: '2'  
}];
```

Choose a delivery option:

- ☒ Monday, October 9
FREE Shipping
- ☐ Thursday, October 5
\$4.99 - Shipping
- ☐ Tuesday, October 3
\$9.99 - Shipping

Choose a delivery option:

- ☐ Monday, October 9
FREE Shipping
- ☒ Thursday, October 5
\$4.99 - Shipping

Quantity: 1 [Update](#) [Delete](#)

14:39:23 / 22:15:56



Create



e a deli

Monday, O

FREE Shipp

Thursday,

99 - Ship

uesday, O

99 - Ship

81

82

83

84

85

86

87

88

89

90

91

92

```
deliveryOptions.forEach((deliveryOption) => {
```

```
  const today = dayjs();
```

```
  const deliveryDate = today.add(
```

```
    deliveryOption.deliveryDays,
```

```
    'days'
```

```
  );
```

```
  const dateString = deliveryDate.format(
```

```
    'dddd, MMMM D'
```

```
  );
```

```
  const priceString = deliveryOption.priceCents
```

```
    === 0
```

delivery option so we can reuse this code

```
const deliveryOptionId = cartItem.deliveryOptionId;
let deliveryOption;

deliveryOptions.forEach((option)=>{
  if(option.id === deliveryOptionId){
    deliveryOption = option;
  }
});

const today = dayjs();
const deliveryDate = today.add(
  deliveryOption.deliveryDays,
  'days'
);
const dateString = deliveryDate.format(
  'dddd, MMMM D'
);
```

- Delivery date: Tuesday, June 21

+ Delivery date: \${dateString}